ECA0806-Analog & Digital Communication

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Experiment-4

MATLAB SIMULATION OF SUPERHETERODYNE RECEIVER FOR AM COMMUNICATION SYSTEM WITH FREQUENCY MIXING AND FILTERING

Program (Matlab Code):

```
% Experiment 4: Simulation of Superheterodyne Receiver (100% Toolbox-Free Version)
clc;
clear;
close all;
% Parameters
fs = 100000;         % Sampling Frequency (Hz)
t = 0:1/fs:0.02;     % Time Vector (s)
fm = 500;            % Message Frequency (Hz)
fc = 10000;          % Carrier Frequency (Hz)
fLO = 14550;         % Local Oscillator Frequency (IF = 4550 Hz)
% Message Signal
m = cos(2 * pi * fm * t);
% Carrier Signal (RF)
c = cos(2 * pi * fc * t);

% Conventional AM Signal
am = (1 + m) .* cos(2 * pi * fc * t);

% Local Oscillator Signal
LO = cos(2 * pi * fLO * t);
% Frequency Mixing
mixed = am .* LO;

% IF Filtering via Frequency Domain (Bypasses Signal Processing Toolbox)
N = length(mixed);
X = fftshift(fft(mixed));
f_axis = linspace(-fs/2, fs/2, N);

% Create an IF bandpass filter mask centered around f_IF = 4550 Hz
f_IF = abs(fLO - fc);
H = zeros(size(f_axis));
passband_idx = (abs(f_axis) >= f_IF - 500) & (abs(f_axis) <= f_IF + 500);
H(passband_idx) = 1;

% Apply filter
X_filtered = X .* H;
IF = real(ifft(ifftshift(X_filtered)));

% Envelope Detection using Rectification (Bypasses hilbert)
rectified = abs(IF);
% Recover Message using a Moving Average Filter (Low-Pass)
```

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```
window_size = 40;
b_lpf = ones(1, window_size) / window_size;
recovered = filter(b_lpf, 1, rectified);
recovered = recovered - mean(recovered); % Remove DC offset
recovered = recovered / max(abs(recovered)) * max(abs(m)); % Scale amplitude
% Display Results in Time Domain Subplots
figure('Name', 'Superheterodyne Receiver Simulation', 'NumberTitle', 'off');

subplot(5, 1, 1);
plot(t, m, 'LineWidth', 1.2);
title('Message Signal');
grid on;
subplot(5, 1, 2);
plot(t, c, 'LineWidth', 1.2);
title('Carrier Signal');
grid on;
subplot(5, 1, 3);
plot(t, am, 'LineWidth', 1.2);
title('AM Signal');
grid on;
subplot(5, 1, 4);
plot(t, IF, 'LineWidth', 1.2);
title('Intermediate Frequency (IF) Signal');
grid on;
subplot(5, 1, 5);
plot(t, recovered, 'r', 'LineWidth', 1.5);
title('Recovered Signal (Output)');
grid on;
```

Output Waveform :

